

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

OBJECT ORIENTED DATA STRUCTURE

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated against them.
- (2) Assume suitable data and illustrate answers with Java code and diagram wherever necessary.

1. (a) Answer the following questions with justification :
What will be the output of the following Java Code ?

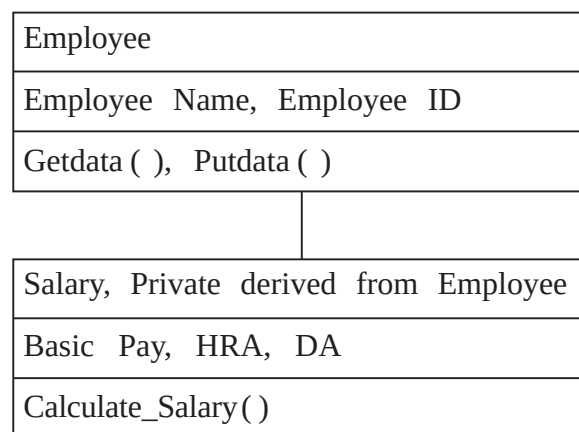
```
(I) class Test1
{
    public static void main(String args[])
    {
        int a = 10 ; int a1 = 4, b = 7 ;
        int result = (true ? a1 & b : a1 | b) > 3 ? 120 : 150 ;
        System.out.println(result) ;
        System.out.println("a ++ = " + (++a + --b)) ;
        System.out.println("b -- = " + (a --)) ;
    }
}

(I) class Left_Shift
{
    public static void main(String args[])
    {
        byte x = 64 ; int i ; byte y ;
        i = x << 2 ;
        y = (byte) (x << 2) ;
        System.out.print(i + " " + y) ;
    }
}
```

2(CO1)

- (b) Explain all features of Object Oriented Programming with example. 3(CO1,2)
- (c) Create a class with the following description : class **Person** having Name, Height and Date of Birth as data members and methods as : input_data (), display_data (), Calculate_Age () method-to calculate age of the person, IsAdult () method-to return true if the person is above 18. Write a Java program to test the class and its methods. 5(CO1,2)
2. (a) Create a class **Clock** which represents a clock (24 hour clock) that keeps the track of time as hours, minutes and seconds. Create a constructor with parameters as hours, minutes and seconds. Include method that allow user to set the time (i.e. to change the hours, minutes and seconds), also include a method display time (int mode) to display the time of the clock and the display of time will depend on mode.
 Mode 1 : indicates the time in 24 hour clock.
 Mode 2 : indicates the time in 12 hour clock.
 Create a class **Clock_demo** which include main () and create object of class Clock and use all methods. 5(CO1,2)
- (b) Create a class **Car** with data members as model_number, limit = 60, speed and fine. Include method input_data () to input information for N cars and method display_data () to print the information of car, also display the message as "You have been fined rupees 1000" if speed greater than limit else display message as "Thank you for driving safely" [Use array of objects]. 5(CO1,2)

3. (a)



- (i) Identify the type of inheritance relationship between the two classes shown in the figure above.

(ii) Define classes to appropriately represent class hierarchy as shown in above figure.

(iii) Use constructors for both classes and display Salary for a particular employee. 2+2+2(CO2)

(b) Write an interface Car having methods :

(i) void changeGear(int newValue)

(ii) void speedUp(int incrementFactor)

(iii) void applyBrakes(int decrementFactor)

Create a class called **HyundaiCreta** having data members speed and gear that implements the above interface and prints the speed and gear after applying above methods. 4(CO2)

4. (a) Why do we use asymptotic notations in algorithm ? Describe commonly used asymptotic notations with their significance. 5(CO3)

(b) Find the time complexity for the given piece of code and elaborate it :

```
(I) class GFG {  
    public static void main(String args[ ])   
    {  
        int i, n = 8 ;  
        for (i = 2 ; i <= n ; i = (int)Math.pow(i, 2))  
        {  
            System.out.println("Hello World") ;  
        }  
    }  
}
```

```
(II) Pseudocode : list_Sum(A, n)  
{  
    total = 0  
    for i = 0 to n - 1  
        sum = sum + a[i]  
    return sum  
}
```

5(CO3)

5. (a) Write Java method or algorithm to implement primitive operations of stack. Convert infix expression :
$$(A + B) * (C * D - E) * F / G$$
to its postfix using stack. [Show all the steps of conversion using stack] 5(CO4)
- (b) Suppose a queue is in circular fashion. It is desired to add to add new items to the queue. Write an algorithm to achieve this, also check whether the queue is full or not. Write another algorithm to delete an element after checking queue empty status. 5(CO4)
6. (a) Consider a singly linked linear list. Write Java method or algorithm :
(i) To create a list using element insertion at the end.
(ii) To reverse the list without creating a new list. 5(CO4)
- (b) Write an algorithm to insert a node between two nodes in doubly linked list. 5(CO4)

